

The Cosmic Microwave Background II

From primordial fluctuations to angular spectra and real data

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Starting point and goal. Lecture I ended with acoustic oscillations of the photon–baryon fluid at last scattering. Before propagating those fluctuations to the observer, we ask what set their unusually simple initial conditions. We then project three-dimensional Fourier modes onto a two-dimensional sky, and derive the angular spectrum C_ℓ . We then learn what changes when cosmological parameters are varied and finish by plotting and fitting the public Planck 2018 temperature data ourselves.

1 Lecture I recap

Lecture I established the physical ingredients that will become the initial sources for today’s angular-spectrum calculation. The Friedmann equation fixes the expanding thermal background. Recombination lowers the free-electron abundance, making the photon mean free path grow and producing a narrow visibility function around last scattering. At linear order, perturbations can be decomposed into independently evolving Fourier modes. Before decoupling, photon pressure and baryon inertia drive acoustic oscillations in the effective temperature

$$X \equiv \Theta_0 + \Psi \tag{1}$$

and in the photon–baryon velocity. Baryon loading shifts the oscillator equilibrium, gravity drives the motion, and photon diffusion damps sufficiently short wavelengths. Lecture II asks how these three-dimensional perturbations appear on a two-dimensional sky and how an experiment measures their statistics.

2 Why inflation?

The motivation here follows the inflation chapter of the Particle Cosmology lecture notes. It is useful to separate the reasons inflation was originally proposed from the properties of perturbations that were measured later. The original background puzzles were the near-flat geometry and the uniform CMB temperature. Detailed CMB measurements subsequently

revealed coherent acoustic phases, approximate scale invariance, and predominantly adiabatic perturbations. Inflation provides one dynamical framework for both sets of facts.

2.1 The flatness problem

Allowing spatial curvature, the Friedmann equation is

$$H^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}. \quad (2)$$

The sign convention for the curvature density parameter varies in the literature. For the present argument only its magnitude matters:

$$|\Omega_k| \equiv \frac{|k|}{a^2 H^2} = \frac{|k|}{(aH)^2}. \quad (3)$$

During radiation domination $H^2 \propto a^{-4}$, giving $|\Omega_k| \propto a^2$; during matter domination $H^2 \propto a^{-3}$, giving $|\Omega_k| \propto a$. Curvature therefore grows relative to the dominant energy density throughout the ordinary decelerating hot-Big-Bang evolution.

Running this evolution backward, the small curvature allowed today requires a much smaller value in the early Universe. Exact flatness is mathematically possible, but without a mechanism that drives the Universe toward flatness it is a finely selected initial condition. Since $|\Omega_k| \propto (aH)^{-2}$, a period during which aH increases instead drives curvature toward zero.

2.2 The horizon problem

The comoving particle horizon is the distance light could have travelled since the beginning of the relevant expansion history,

$$\chi_H(a) = \int_{a_i}^a \frac{d\tilde{a}}{\tilde{a}^2 H(\tilde{a})}. \quad (4)$$

In a purely radiation- and matter-dominated history this distance is finite. At last scattering it is of order a few hundred comoving megaparsecs, whereas the comoving distance from us to the last-scattering surface is roughly 14 Gpc. The causal scale therefore covers only an angular region of order a degree. Many widely separated regions on the observed CMB sky could not have exchanged signals before recombination in that history, yet their mean temperatures agree to about one part in 10^5 .

The problem is not that photons from these regions cannot reach us: they clearly do. It is that the regions could not communicate with one another early enough to establish nearly identical thermal conditions. A causal explanation requires a different earlier expansion history or finely correlated initial conditions.

2.3 Accelerated expansion addresses both background puzzles

Inflation is a period of accelerated expansion, so both

$$\dot{a} > 0, \ddot{a} > 0. \quad (5)$$

Since $aH = \dot{a}$, the general condition for the magnitude of the comoving Hubble radius to shrink is

$$\frac{d}{dt} \frac{1}{|aH|} < 0 \quad \Longleftrightarrow \quad \dot{a} \ddot{a} > 0. \quad (6)$$

This also allows an accelerated contracting branch, with $\dot{a} < 0$ and $\ddot{a} < 0$; inflation refers specifically to the expanding branch. The comoving Hubble radius therefore shrinks during inflation. A comoving wavelength can begin below this scale, cross outside it during inflation, and re-enter when the comoving Hubble radius grows again during radiation or matter domination. The whole region visible to us can descend from one much smaller patch with a common causal past. At the same time, $|\Omega_k| \propto (aH)^{-2}$ is dynamically suppressed.

The Hubble radius and the particle horizon are not identical concepts. The former is the expansion timescale expressed as a length; the latter is an integrated causal distance. During the standard decelerating eras they are of the same order. Inflation changes the lower limit and the expansion history in the particle-horizon integral, allowing the shared causal past that is absent from an extrapolation of the hot Big Bang alone.

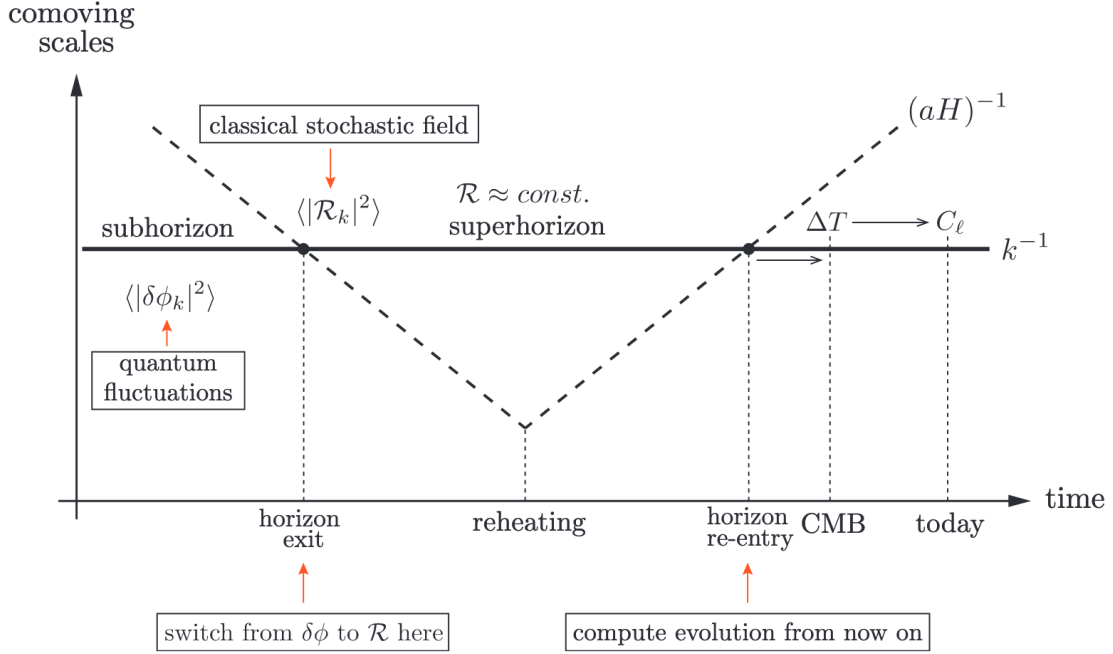


Figure 1: A fixed comoving wavelength begins below the comoving Hubble scale, exits during inflation, and re-enters during the later decelerating expansion.

2.4 Later CMB evidence concerns the fluctuations

Lecture I derived acoustic motion mode by mode. The observed sequence of sharp peaks requires the modes to enter the horizon in a common temporal phase. Random phases would average the oscillations into a broad, smooth spectrum. The coherence is seen not only in temperature but also in the relative phases of temperature and polarization.

The inferred primordial curvature power spectrum is close to scale invariant, and the

perturbations are predominantly adiabatic. The measured spectrum is well described by

$$\Delta_{\mathcal{R}}^2(k) = A_s \left(\frac{k}{k_0} \right)^{n_s-1}, \quad n_s \simeq 1. \quad (7)$$

Thus the power per logarithmic interval varies only slowly with scale. In the simplest single-field picture, quantum fluctuations begin on subhorizon scales, are stretched beyond the Hubble radius, and become the classical curvature perturbation $\mathcal{R}$. A single clock also selects adiabatic initial conditions and a common growing mode, explaining the phase coherence.

We will not derive the quantum generation of $\mathcal{R}$ here. For the CMB calculation we need its statistical spectrum and the fact that subsequent linear evolution is deterministic for each mode. This is precisely the separation between primordial statistics and transfer functions used below.

3 From last scattering to the observer

3.1 What was present at recombination?

For each Fourier mode $\mathbf{k}$, Lecture I identified three important quantities at the last-scattering time τ_* :

- the photon temperature monopole $\Theta_0 = \delta_\gamma/4$;
- the baryon velocity $\mathbf{v}_b$, approximately equal to the photon velocity in tight coupling;
- the metric potentials Φ and Ψ in conformal Newtonian gauge,

$$ds^2 = a^2(\tau) \left[-(1 + 2\Psi)d\tau^2 + (1 - 2\Phi)d\mathbf{x}^2 \right].$$

Here Θ_0 denotes the angular monopole of the photon-temperature perturbation for a Fourier mode. It is unrelated to the symbol T_0 for the mean CMB temperature today. The photon-baryon density is at an extremum when its velocity is approximately zero, and vice versa. Density and velocity contributions will therefore appear out of phase.

3.2 Gravitational and scattering contributions

For orientation, imagine instantaneous recombination. Integration of the photon geodesic and the Thomson-scattering source gives the observed fractional temperature in direction $\hat{\mathbf{n}}$,

$$\Theta(\hat{\mathbf{n}}) = \underbrace{\left[\frac{1}{4}\delta_\gamma + \Psi \right]_*}_{\text{Sachs-Wolfe}} - \underbrace{\hat{\mathbf{n}} \cdot \mathbf{v}_b|_*}_{\text{Doppler}} + \underbrace{\int_{\tau_*}^{\tau_0} d\tau (\dot{\Phi} + \dot{\Psi})}_{\text{integrated Sachs-Wolfe}}. \quad (8)$$

Dots denote derivatives with respect to conformal time.

The three pieces have distinct origins:

1. $\delta_\gamma/4$ is an intrinsic temperature perturbation.
2. Ψ_* gravitationally redshifts a photon climbing out of a potential well. Together these form the ordinary Sachs–Wolfe contribution.
3. Motion of the last scatterers produces a line-of-sight Doppler shift.
4. If the potentials evolve while a photon travels, it gains or loses additional energy. This is the integrated Sachs–Wolfe (ISW) effect.

Potentials are nearly constant during matter domination. They evolve near radiation–matter equality and again when dark energy becomes important. Early ISW affects the acoustic region; late ISW mainly affects large angular scales.

3.3 The more accurate line-of-sight solution

Recombination is not instantaneous. With optical depth $\tilde{\tau}$ and visibility $g = -\dot{\tilde{\tau}}e^{-\tilde{\tau}}$, all local temperature sources can be collected in a source function $S_T(k, \tau)$. To connect this notation to Eq. (8), write $\chi(\tau) = \tau_0 - \tau$. Schematically, projection into multipoles gives

$$\Theta_\ell(k) \simeq \int_0^{\tau_0} d\tau \left[gX j_\ell(k\chi) + gv_b j'_\ell(k\chi) + e^{-\tilde{\tau}}(\dot{\Phi} + \dot{\Psi})j_\ell(k\chi) + \cdots \right]. \quad (9)$$

The first two terms are the finite-width versions of the last-scattering emission and Doppler terms, while the third is accumulated along the photon path. The ellipsis includes polarization and other terms in the exact linear treatment; detailed signs and velocity factors depend on Fourier conventions. Integrating the Doppler term by parts moves the derivative from j_ℓ into its coefficient. All coefficients of j_ℓ can then be packaged in $S_T(k, \tau)$, giving the exact linear form

$$\Theta_\ell(k) = \int_0^{\tau_0} d\tau S_T(k, \tau) j_\ell[k(\tau_0 - \tau)]. \quad (10)$$

S_T is therefore a compact organization of the visibility-weighted monopole, potential, velocity, polarization, and ISW terms, rather than a new physical effect. The spherical Bessel function j_ℓ performs the geometric projection from a Fourier wavelength to an angular multipole. CAMB and CLASS exploit this line-of-sight organization rather than evolving every photon multipole until today.

Check your understanding. Why does one Fourier mode contribute to a range of ℓ rather than one exact ℓ ? Projection through $j_\ell(k\chi)$ has finite width, and real last scattering also has finite radial width. Nevertheless it is useful to remember $\ell \sim k\chi_*$.

4 Temperature as a random field on the sky

4.1 Why spherical harmonics?

The CMB temperature is a scalar field on a sphere. The analogue of a Fourier expansion is a spherical-harmonic expansion,

$$\Theta(\hat{\mathbf{n}}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\hat{\mathbf{n}}), \quad (11)$$

with inverse

$$a_{\ell m} = \int d\Omega_{\hat{\mathbf{n}}} Y_{\ell m}^*(\hat{\mathbf{n}}) \Theta(\hat{\mathbf{n}}). \quad (12)$$

The integers have a geometric meaning:

- $\ell = 0$ is the monopole, the mean CMB temperature;
- $\ell = 1$ is the dipole, dominated by our peculiar velocity;
- increasing ℓ describes progressively smaller angular structure, roughly $\theta \sim \pi/\ell$;
- for every ℓ there are $2\ell + 1$ values of m .

4.2 Statistical isotropy and the angular spectrum

We have only one CMB sky, but theoretical initial conditions are stochastic. Angular brackets denote an ensemble average over hypothetical realizations with the same cosmological parameters. Statistical isotropy says that no orientation is preferred. It implies

$$\langle a_{\ell m} a_{\ell' m'}^* \rangle = C_{\ell} \delta_{\ell \ell'} \delta_{m m'}. \quad (13)$$

Thus all $2\ell + 1$ modes at fixed ℓ have the same variance. For a statistically isotropic Gaussian field, the set of C_{ℓ} contains all two-point information.

The real-space angular correlation function is

$$\langle \Theta(\hat{\mathbf{n}}) \Theta(\hat{\mathbf{n}}') \rangle = \sum_{\ell} \frac{2\ell + 1}{4\pi} C_{\ell} P_{\ell}(\hat{\mathbf{n}} \cdot \hat{\mathbf{n}}'). \quad (14)$$

This follows from the addition theorem

$$\sum_m Y_{\ell m}(\hat{\mathbf{n}}) Y_{\ell m}^*(\hat{\mathbf{n}}') = \frac{2\ell + 1}{4\pi} P_{\ell}(\hat{\mathbf{n}} \cdot \hat{\mathbf{n}}').$$

4.3 Estimating the spectrum from one sky

On a complete, noiseless sky an unbiased estimator is

$$\hat{C}_{\ell} = \frac{1}{2\ell + 1} \sum_{m=-\ell}^{\ell} |a_{\ell m}|^2. \quad (15)$$

For Gaussian $a_{\ell m}$,

$$\boxed{\text{Var}(\widehat{C}_\ell) = \frac{2}{2\ell + 1} C_\ell^2.} \quad (16)$$

This is *cosmic variance*. Even a perfect instrument cannot provide additional independent m modes on the largest scales. At high ℓ there are many modes and the fractional uncertainty becomes smaller.

Observers usually plot power per logarithmic interval in angular scale,

$$\boxed{D_\ell \equiv \frac{\ell(\ell + 1)}{2\pi} C_\ell.} \quad (17)$$

The measured D_ℓ^{TT} has units of μK^2 when dimensional temperature fluctuations rather than Θ are used. Since $\delta T = T_0 \Theta$, the dimensionless and dimensional spectra are related by

$$\boxed{C_\ell^{TT, \mu\text{K}^2} = T_0^2 C_\ell^{\Theta\Theta},} \quad (18)$$

provided T_0 is expressed in μK . The same conversion applies to D_ℓ .

Key result. The $a_{\ell m}$ describe our particular sky. The C_ℓ are ensemble variances predicted by a model. We estimate them by averaging over the $2\ell + 1$ values of m , which is why cosmic variance remains.

5 Deriving the temperature angular spectrum

We now connect the primordial three-dimensional perturbations to the observed two-dimensional spectrum, following the notation and derivation in the Particle Cosmology notes.

5.1 Separate initial conditions from deterministic evolution

Linear evolution allows us to factor the temperature contribution of a Fourier mode into the initial curvature perturbation $\mathcal{R}_i(\mathbf{k})$ and a transfer function. One may write

$$\Theta(\hat{\mathbf{n}}) = \sum_\ell i^\ell (2\ell + 1) \int \frac{d^3 k}{(2\pi)^3} \Theta_\ell(k) \mathcal{R}_i(\mathbf{k}) P_\ell(\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}). \quad (19)$$

The transfer function $\Theta_\ell(k)$ contains all linear evolution after the initial conditions: acoustic oscillations, baryon loading, gravitational driving, diffusion, recombination, Doppler and ISW effects, and projection.

For instantaneous recombination, its schematic form is

$$\Theta_\ell(k) = \tilde{A}(k) j_\ell(k\chi_*) - \tilde{B}(k) j'_\ell(k\chi_*), \quad (20)$$

The prime on j'_ℓ means differentiation with respect to its complete argument $k\chi_*$; it is not a conformal-time derivative. Here $\tilde{A}$ contains the monopole, potentials, and ISW contribution per unit $\mathcal{R}_i$, while $\tilde{B}$ contains the baryon velocity per unit $\mathcal{R}_i$. The full result is Eq. (10).

5.2 The primordial power spectrum

Statistical homogeneity and isotropy define

$$\langle \mathcal{R}_i(\mathbf{k}) \mathcal{R}_i(\mathbf{k}') \rangle = (2\pi)^3 \delta(\mathbf{k} + \mathbf{k}') \frac{2\pi^2}{k^3} \Delta_{\mathcal{R}}^2(k). \quad (21)$$

For the simplest inflationary initial conditions,

$$\Delta_{\mathcal{R}}^2(k) = A_s \left(\frac{k}{k_0} \right)^{n_s-1}. \quad (22)$$

A_s fixes the primordial amplitude and n_s its scale dependence.

5.3 Compute the two-point function

Insert Eq. (19) twice into $\langle \Theta(\hat{\mathbf{n}}) \Theta(\hat{\mathbf{n}}') \rangle$. The delta function in Eq. (21) removes one momentum integral. The angular integral uses

$$\int d\hat{\mathbf{k}} P_\ell(\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}) P_{\ell'}(\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}') = \frac{4\pi}{2\ell+1} P_\ell(\hat{\mathbf{n}} \cdot \hat{\mathbf{n}}') \delta_{\ell\ell'}. \quad (23)$$

The result is

$$\langle \Theta(\hat{\mathbf{n}}) \Theta(\hat{\mathbf{n}}') \rangle = \sum_{\ell} \frac{2\ell+1}{4\pi} \left[4\pi \int d\ln k \Theta_{\ell}^2(k) \Delta_{\mathcal{R}}^2(k) \right] P_{\ell}(\hat{\mathbf{n}} \cdot \hat{\mathbf{n}}'). \quad (24)$$

Comparison with Eq. (14) identifies

$$C_{\ell}^{TT} = 4\pi \int d\ln k \Theta_{\ell}^2(k) \Delta_{\mathcal{R}}^2(k). \quad (25)$$

For any two linear observables $X, Y \in \{T, E\}$,

$$C_{\ell}^{XY} = 4\pi \int d\ln k \Theta_{\ell}^X(k) \Theta_{\ell}^Y(k) \Delta_{\mathcal{R}}^2(k). \quad (26)$$

Key result. Primordial physics is encoded in $\Delta_{\mathcal{R}}^2(k)$. Everything that happens between the initial conditions and the observed sky is encoded in the transfer functions $\Theta_{\ell}^X(k)$. Their convolution is the observable C_{ℓ}^{XY} .

5.4 A useful large-scale check

On scales larger than the sound horizon at recombination, the ordinary Sachs–Wolfe relation gives $\Theta \simeq \mathcal{R}/5$. The transfer function is approximately

$$\Theta_{\ell}(k) \simeq \frac{1}{5} j_{\ell}(k\chi_*) \quad (27)$$

For a scale-invariant primordial spectrum, the identity

$$\int_0^\infty \frac{dk}{k} j_\ell^2(k\chi_*) = \frac{1}{2\ell(\ell+1)}$$

implies

$$\frac{\ell(\ell+1)}{2\pi} C_\ell^{TT} \simeq \frac{A_s}{25}, \quad (28)$$

approximately constant. This is the Sachs–Wolfe plateau. Acoustic dynamics breaks the plateau into peaks at smaller angular scales.

6 How to read a CMB spectrum

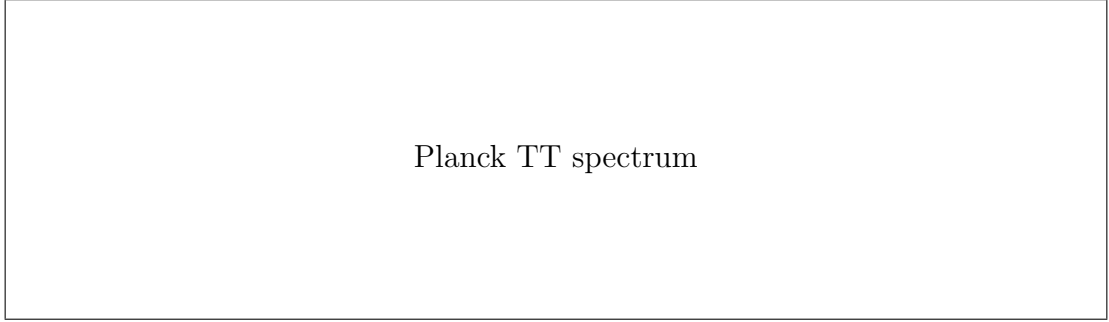


Figure 2: Planck temperature bandpowers and a best-fit Λ CDM spectrum. The vertical axis is $D_\ell^{TT} = \ell(\ell+1)C_\ell^{TT}/(2\pi)$. Figure retained from the Particle Cosmology course material, adapted there from Planck 2018.

The observed spectrum is a compressed history of the Universe:

- **Low ℓ :** ordinary and integrated Sachs–Wolfe effects; cosmic variance is large.
- **Acoustic region:** coherent photon–baryon oscillations and gravitational driving.
- **High ℓ :** photon diffusion, finite recombination width, lensing, and foregrounds.

The six parameters of minimal flat Λ CDM are commonly chosen as

$$\{\omega_b, \omega_c, \theta_s, \tau_{\text{reio}}, A_s, n_s\}, \quad (29)$$

where $\omega_i \equiv \Omega_i h^2$ are physical densities.

Change	Main signature, with other parameters held fixed where possible
$A_s \uparrow$	Raises primordial power and hence primary TT/TE/EE; also increases lensing.
$n_s \uparrow$	Tilts the spectrum, increasing small-scale power relative to large-scale power around the pivot.
$\omega_b \uparrow$	Adds baryon inertia: odd compression peaks rise relative to even rarefaction peaks; c_s decreases.
$\omega_c \uparrow$	Moves equality earlier and changes potential decay, radiation driving, and the peak envelope.
$\theta_s \uparrow$	Increases the apparent angular sound horizon and moves the acoustic pattern toward lower ℓ .
$\tau_{\text{reio}} \uparrow$	Suppresses primary power at $\ell \gtrsim 20$ by $e^{-2\tau_{\text{reio}}}$ and creates low- ℓ EE power.
$N_{\text{eff}} \uparrow$	Speeds pre-recombination expansion, changes r_s/r_D , and produces a free-streaming phase shift.
$\sum m_\nu \uparrow$	Changes late-time expansion and suppresses structure growth and CMB lensing.

6.1 Peak positions: a standard ruler

The characteristic acoustic scale is

$$\theta_s = \frac{r_s(\tau_*)}{\chi_*}, \quad \ell_A \simeq \frac{\pi}{\theta_s}. \quad (30)$$

Pre-recombination physics predicts r_s ; the expansion history between us and last scattering determines χ_* . The CMB measures their ratio extremely precisely. Different late-time cosmologies can preserve the same ratio, producing the geometric degeneracy.

6.2 Peak heights: competing physical effects

The ratio of odd to even peaks measures baryon loading. The overall envelope measures potential decay around equality and thus the matter-to-radiation ratio. The damping tail compares the diffusion scale with the sound horizon and is sensitive to the expansion rate and recombination microphysics.

No parameter changes exactly one feature. A proper analysis predicts the full set of spectra for every parameter point and varies foreground and instrumental nuisance parameters simultaneously.

6.3 Polarization supplies phase and geometry information

Thomson scattering generates linear polarization only if the incident radiation has a quadrupole. A small quadrupole develops as photons begin to free-stream near recombination and again at reionization. The Stokes fields Q and U combine into parity-even E and parity-odd B polarization.

Scalar perturbations produce T and E but no primordial B at linear order. The scalar spectra are therefore TT, TE, and EE. Since velocity is out of phase with density, E-mode peaks and TE sign changes provide a strong test of coherent acoustic oscillations. Reionization creates a low- ℓ EE bump and allows τ_{reio} to be separated from the high- ℓ amplitude combination $A_s e^{-2\tau_{\text{reio}}}$.

Weak gravitational lensing remaps the sky,

$$X(\hat{\mathbf{n}}) \longrightarrow X(\hat{\mathbf{n}} + \nabla\phi), \quad (31)$$

smooths acoustic peaks, converts E into B, and generates a connected four-point function from which the lensing potential ϕ can be reconstructed.

Check your understanding. Why does measuring EE help if TT already has high signal-to-noise? It observes different source combinations and phases, breaks parameter degeneracies, and provides powerful consistency tests against foregrounds and instrumental systematics.

7 How a real experiment obtains bandpowers

The points in Fig. 2 are not obtained by simply transforming one perfect temperature map. The actual chain contains several inference problems.

7.1 Time-ordered detector data

A scanning polarized detector records a timestream that can be written schematically as

$$d_t = g_t [I_p + Q_p \cos(2\gamma_t) + U_p \sin(2\gamma_t)] + n_t + s_t. \quad (32)$$

The scalar d_t is one detector datum at time-sample index t . Pointing selects $p = p(t)$, and γ_t is the detector orientation. I_p is total intensity; Q_p and U_p describe linear polarization relative to a chosen pair of axes in that sky pixel. The factor g_t is calibration, n_t is noise, and s_t represents instrumental systematics. Beams, time constants, glitches, pointing, gain, and bandpasses must be characterized.

7.2 Map-making

In matrix notation,

$$\mathbf{d} = \mathbf{A}\mathbf{m} + \mathbf{n}, \quad (33)$$

where the boldface data vector stacks all samples, $\mathbf{d} = (d_1, \dots, d_{N_t})^T$. The pointing matrix $\mathbf{A}$ connects those samples to the map vector $\mathbf{m}$, which contains (I_p, Q_p, U_p) for each pixel. For Gaussian noise covariance $\mathbf{N}$, the generalized least-squares map is

$$\hat{\mathbf{m}} = (\mathbf{A}^T \mathbf{N}^{-1} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{N}^{-1} \mathbf{d}. \quad (34)$$

Large experiments solve this iteratively and use data splits and difference maps to look for residual systematics.

7.3 Foreground separation

The sky depends on observing frequency,

$$m_\nu(\hat{\mathbf{n}}) = a_{\text{CMB}}(\hat{\mathbf{n}}) + \sum_c f_\nu^c(\boldsymbol{\beta}_c, \hat{\mathbf{n}}) a_c(\hat{\mathbf{n}}) + n_\nu(\hat{\mathbf{n}}). \quad (35)$$

Galactic synchrotron dominates polarized low frequencies; thermal dust dominates high frequencies. Temperature also contains free-free and spinning-dust emission, radio and infrared sources, the cosmic infrared background, and thermal and kinetic Sunyaev–Zel’dovich signals. Multiple frequency channels are essential because the CMB has a known blackbody frequency dependence while foregrounds differ.

7.4 Cut-sky spectrum estimation

A Galactic and point-source mask destroys exact spherical-harmonic orthogonality and couples multipoles. A beam suppresses small angular scales, filtering removes modes, and noise adds power. Schematically,

$$\langle \tilde{C}_\ell \rangle = \sum_{\ell'} M_{\ell\ell'} B_{\ell'}^2 F_{\ell'} C_{\ell'} + \tilde{N}_\ell. \quad (36)$$

M is the mask mode-coupling matrix, B_ℓ the beam window, and F_ℓ the filter transfer function. Cross-spectra between maps with independent noise avoid the mean auto-spectrum noise bias. Nearby multipoles are then grouped into bandpowers.

For a quick uncertainty estimate,

$$\text{Var}(\hat{C}_\ell) \simeq \frac{2}{(2\ell + 1)f_{\text{sky}}} (C_\ell + N_\ell)^2. \quad (37)$$

This is useful intuition, but real bandpower covariance matrices include mode coupling, nonuniform noise, beam uncertainty, and foreground correlations.

7.5 Likelihood and parameter posterior

Finally,

$$P(\boldsymbol{\theta}, \boldsymbol{\eta} \mid \mathbf{d}) \propto \mathcal{L}[\mathbf{d} \mid C_\ell(\boldsymbol{\theta}), \boldsymbol{\eta}] P(\boldsymbol{\theta}) P(\boldsymbol{\eta}), \quad (38)$$

where $\boldsymbol{\theta}$ are cosmological parameters and $\boldsymbol{\eta}$ nuisance parameters. CAMB or CLASS supplies $C_\ell(\boldsymbol{\theta})$; a sampler explores the posterior. Low multipoles require non-Gaussian or map-based likelihood treatments, while suitable high- ℓ bandpowers admit accurate approximations.

Key result. A published spectrum is a calibrated, foreground-cleaned, beam- and mask-corrected statistical estimate with a covariance and nuisance model. It is not the raw squared transform of a picture of the sky.

8 Hands-on: fit the real Planck TT spectrum

The accompanying notebook is

[Open the Planck TT exercise directly in Google Colab.](#)

The repository also contains the downloadable file `Planck_TT_data_analysis.ipynb`. No local installation is required when using the direct Colab link.

The notebook downloads the official Planck 2018 (PR3) binned TT spectrum from the Planck archive mirror. The table contains

$$\{\ell_{\text{eff}}, D_{\ell}^{\text{data}}, \sigma_{-}, \sigma_{+}, D_{\ell}^{\text{best fit}}\}. \quad (39)$$

Students first reproduce the familiar plot of the measured acoustic peaks. They then fit the deliberately simple model

$$D_{\ell}^{\text{model}}(A, \alpha) = A D_{\alpha\ell}^{\text{template}}. \quad (40)$$

The two phenomenological parameters isolate two ideas discussed above:

- A rescales the vertical amplitude, loosely resembling a change in $A_s e^{-2\tau_{\text{reio}}}$;
- α dilates the horizontal multipole axis, loosely resembling a change in the angular sound scale θ_s .

For the classroom fit, asymmetric errors are averaged and treated as independent Gaussian uncertainties,

$$\chi^2(A, \alpha) = \sum_b \frac{[D_b^{\text{data}} - D_b^{\text{model}}(A, \alpha)]^2}{\sigma_b^2}. \quad (41)$$

The notebook plots the best fit, normalized residuals, and approximate 68% and 95% contours in the (A, α) plane.

8.1 What this fit teaches—and what it does not

It teaches that a model becomes a curve in data space, uncertainties define a likelihood, parameters can be correlated, residuals diagnose mismatches, and acoustic positions are measured far more sharply than one might guess by eye.

It is not a cosmological parameter analysis. The public bandpowers have correlated errors; the low- ℓ likelihood is not Gaussian; foreground and calibration parameters have already entered the published spectrum; and the template was itself obtained from a Planck best fit. Research inference instead recomputes TT, TE, and EE with an Einstein–Boltzmann solver and evaluates the official likelihood including its covariance and nuisance model.

Check your understanding. Why should the notebook find $A \simeq 1$ and $\alpha \simeq 1$? Because the template was obtained by fitting essentially these data with a much more complete model. The exercise tests our simplified compression of that comparison.

9 Outlook: current instruments and future information

For a measured sky fraction f_{sky} , the approximate spectrum uncertainty

$$\sigma(\hat{C}_\ell) \simeq \sqrt{\frac{2}{(2\ell+1)f_{\text{sky}}}} (C_\ell + N_\ell) \quad (42)$$

organizes the experimental outlook. Better sensitivity and angular resolution lower N_ℓ and make additional polarization and small-scale modes useful. More sky raises the number of independent modes. Once $N_\ell \ll C_\ell$, however, that spectrum on that sky is cosmic-variance limited.

Modern CMB bolometers can approach the photon-noise floor set by emission from the sky, atmosphere, and telescope. This photon noise contains both Poisson fluctuations in the arrival counts and, for thermal radiation, correlations between photon arrivals. The latter contribution is called wave noise or bunching noise. If detector noises are approximately independent,

$$\text{NET}_{\text{array}} \simeq \frac{\text{NET}_{\text{det}}}{\sqrt{N_{\text{det}}}}, \quad \text{mapping speed} \propto N_{\text{det}}. \quad (43)$$

This is why the long-running sensitivity trend has depended strongly on putting more background-limited detectors on the sky. Detector yield, atmosphere, correlated noise, calibration, beams, and observing efficiency limit the ideal scaling.

The connection to mode counting is direct:

$$N_{\text{modes}} = f_{\text{sky}} \sum_{\ell=2}^{\ell_{\text{max}}} (2\ell+1) \simeq f_{\text{sky}} \ell_{\text{max}}^2. \quad (44)$$

Future constraints improve when lower noise, broader sky coverage, or better control of systematics and foregrounds makes more of these modes usable. Frequency channels do not supply independent copies of one CMB mode. Instead, they separate the CMB from synchrotron, Galactic dust, the cosmic infrared background, and other emission with different spectra.

9.1 Complementary experiments

The Simons Observatory combines a 6-m Large Aperture Telescope for arcminute scales with Small Aperture Telescopes optimized for degree-scale polarization. Its expanded programme includes two SO:UK mid-frequency SATs and an SO:JP-supported low-frequency SAT. Ten-year forecasts for the full six-SAT array give $\sigma(r) \simeq (0.7\text{--}1.4) \times 10^{-3}$ across optimistic and conservative scenarios, assuming 70% delensing in the quoted n_s - r forecasts.

CCAT/FYST extends ground-based coverage to submillimetre frequencies, where Galactic dust and the cosmic infrared background are bright and where thermal SZ, dusty-galaxy, and line-intensity signals become accessible. LiteBIRD instead targets full-sky, large-angular-scale polarization from space with 15 bands spanning 34–448 GHz and a mission requirement of total uncertainty $\delta r < 10^{-3}$. Their differing angular scales and frequencies make the programmes complementary rather than interchangeable.

9.2 The CMB as a backlight and a non-Gaussian field

The primary CMB is also a precisely characterized backlight. Matter between last scattering and us lenses it, ionized gas scatters it, and hot electrons shift photon frequencies through the Sunyaev–Zel’dovich effect. These secondary changes encode the distribution and thermodynamics of late-time structure.

For a Gaussian sky the two-point function contains all statistical information. Connected higher-order correlations, beginning with the bispectrum, probe additional physics. Primordial non-Gaussianity can test interactions and additional fields during inflation. Secondary non-Gaussianity is generated later by lensing, SZ signals, galaxies, and foregrounds; it is both a contaminant to primordial searches and a valuable observable in its own right. For example, the compressed statistic

$$C_L^{g^2 T_\nu} \quad (45)$$

is a galaxy–galaxy–temperature bispectrum: g traces a low-redshift galaxy field and T_ν is a temperature map at frequency ν . Its frequency dependence helps separate thermal SZ, cosmic-infrared-background, and self-pair contributions. A detection of this statistic is therefore a detection of secondary, late-time non-Gaussianity rather than primordial non-Gaussianity.

End-of-lecture synthesis

$$\boxed{\Delta_{\mathcal{R}}^2(k) \xrightarrow[\text{Einstein–Boltzmann evolution}]{\boldsymbol{\theta}} C_\ell^{XY} \xrightarrow[\text{beam, mask, foregrounds}]{\text{experiment}} \hat{C}_b \xrightarrow[\text{likelihood}]{} P(\boldsymbol{\theta} \mid \text{data}).}$$

The CMB is powerful because the forward calculation is precise and the acoustic pattern is highly structured. The observation is equally impressive: a faint, frequency-dependent, beam-convolved sky is turned into constraints on physics from recombination, the primordial Universe, and the growth of structure.

Notation guide

Symbol	Meaning
$\Theta(\hat{\mathbf{n}}), a_{\ell m}$	temperature field and its spherical-harmonic coefficients
C_ℓ, D_ℓ	angular spectrum and $\ell(\ell+1)C_\ell/(2\pi)$
$\mathcal{R}_i(\mathbf{k}), \Delta_{\mathcal{R}}^2(k)$	initial comoving curvature perturbation and dimensionless power
$\Theta_\ell^X(k)$	radiation transfer function for observable X
Φ, Ψ	conformal-Newtonian-gauge metric potentials
$\tilde{\tau}, g(\tau)$	optical depth and visibility function
r_s, χ_*, θ_s	sound horizon, distance to last scattering, and their angular ratio
ω_b, ω_c	physical baryon and cold-dark-matter densities $\Omega_i h^2$
$\boldsymbol{\theta}, \boldsymbol{\eta}$	cosmological and nuisance parameters

Source and scope note

The line-of-sight result, angular statistics, transfer-function derivation, and C_ℓ normalization follow the Particle Cosmology lecture notes, especially Lectures 2, 4, and 5. The observation and likelihood sections expand the bridge to data that was implicit in those notes. The notebook uses the official public Planck 2018 binned TT product. Its template-dilation fit is explicitly pedagogical and does not reproduce the official Planck likelihood.